

Lecture 6 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 6 introduces hypothesis testing as a way to decide whether data contradict, support, or fail to resolve a null model. The lecture starts by distinguishing raw and normalized effect sizes, then compares Savage-Dickey, Bayes factors, and HDI/ROPE tests as Bayesian approaches to hypothesis testing. It then introduces significance, type I and type II errors, power, and pre-hoc and post-hoc power analysis. The lecture ends by connecting these ideas to frequentist tests and to a two-group hospital example using PCI and non-PCI groups.

Lecture Sections

6.1 — Effect sizes

This section introduces the null model and defines effect size as the difference between reality and the null hypothesis. Raw effect sizes are expressed in the original units of the data and depend on domain knowledge. Normalized effect sizes are designed to be comparable across experiments or domains, but they can hide practical meaning.

New Definitions and Terms

- **Null model** — a default or naïve model, often assuming no effect, no difference, no relationship, or no influence
- **Null hypothesis** — the hypothesis that the null model is true
- **Effect size** — the difference between reality and the null hypothesis
- **Raw effect size** — an effect size expressed in the actual units of the data
- **Normalized effect size** — an effect-size statistic designed to be comparable across experiments and domains
- **True effect size** — the effect size in reality
- **Measured effect size** — the effect size estimated from data
- **Cohen's h** — a normalized effect size for comparing proportions or probabilities

Formulas

$$h = 2 \arcsin \sqrt{\theta} - 2 \arcsin \sqrt{\theta_0}$$

Cohen's h transforms a difference between probabilities into a normalized effect-size scale.

Figure

Many possible effect sizes

Effect-size scale	Formula	Commonly used for
Risk difference / proportion difference	$\Delta = \theta - \theta_0$	Probabilities far from 0 or 1
Risk ratio / relative risk	$RR = \frac{\theta}{\theta_0}$	Small probabilities, clinical applications
Odds ratio	$OR = \frac{\theta}{1-\theta} / \frac{\theta_0}{1-\theta_0}$	Probabilities near 0 or 1, statistics
Log odds ratio	$\log OR$	Transformation of odds ratio to additive domain
Cohen's (h)	$2 \arcsin \sqrt{\theta} - 2 \arcsin \sqrt{\theta_0}$	Normalized effect size

Figure 1: Slide 8 — Many possible effect sizes

6.2 — Hypothesis tests

This section defines hypothesis testing as a way to evaluate whether the null hypothesis remains credible after seeing the data. The key contrast is between effect size and evidence against the null: an estimated effect size can be large or small regardless of sample size, but a hypothesis test depends strongly on the amount of data. Savage-Dickey is introduced as a Bayesian density-ratio method for measuring evidence for or against a reference value.

New Definitions and Terms

- **Hypothesis test** — a procedure for deciding whether data support rejecting, accepting, or withholding judgment about a hypothesis
- **Savage-Dickey density ratio** — the ratio of posterior and prior density at a reference value
- **Reference value** — the parameter value being tested, usually the null value
- **Evidence against the null** — data making the null value less credible than it was under the prior
- **Evidence for the null** — data making the null value more credible than it was under the prior

Formulas

$$BF_{01} = \frac{p(\theta_0 | y)}{p(\theta_0)}$$

The Savage-Dickey density ratio compares posterior and prior density at the null value. A larger posterior density at the null value supports the null; a smaller posterior density supports rejecting it.

Figure

The Savage-Dickey density ratio

- Ratio of prior and posterior density at a reference value
- Measures evidence in data for or against that value

$$BF_{01} = \frac{p(\theta_{\text{ref}} | y)}{p(\theta_{\text{ref}})}$$

```
az.plot_bf(idata, var_name="theta")
```

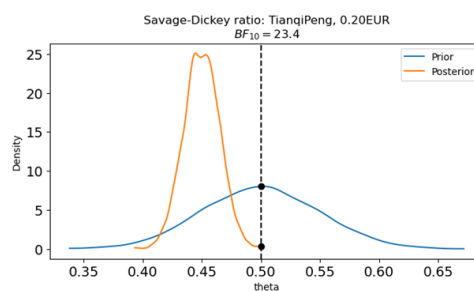


Figure 2: Slide 14 — The Savage-Dickey density ratio

Savage-Dickey depends on effect size and sample size

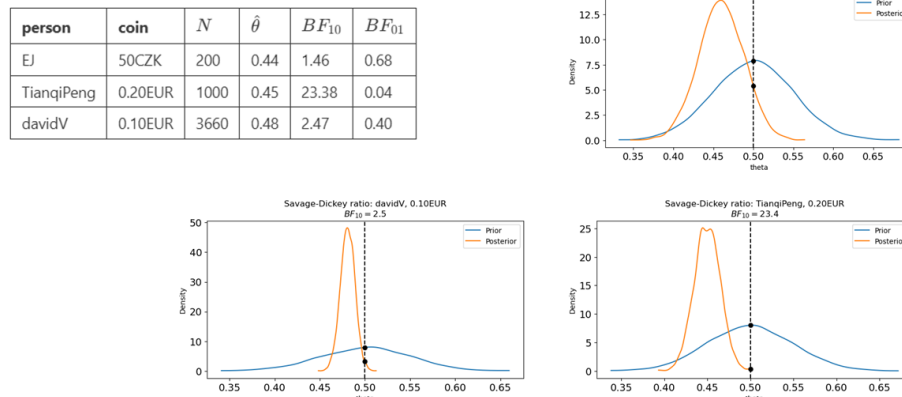


Figure 3: Slide 16 — Savage-Dickey depends on effect size and sample size

6.3 — Baye's factor

This section places Bayes factors inside posterior odds. A Bayes factor is the part of the posterior odds that comes from the data, after separating out the prior odds between models. The lecture emphasizes that Savage-Dickey is a Bayes factor only under specific conditions, and that Bayes factors can be difficult to calculate and interpret.

New Definitions and Terms

- **Posterior odds** — the relative posterior probability of two models
- **Prior odds** — the relative prior probability of two models before seeing data
- **Bayes factor** — the part of the posterior odds contributed by the data
- **Marginal likelihood** — the likelihood of the data under a model after averaging over that model's parameters
- **Point null** — a null hypothesis that assigns one exact parameter value to the null model

Formulas

$$\frac{p(m_1 | y)}{p(m_2 | y)} = \frac{p(y | m_1) p(m_1)}{p(y | m_2) p(m_2)}$$

Posterior odds equal the Bayes factor times the prior odds.

$$p(y | m) = \int d\theta p(y | \theta, m) p(\theta | m)$$

The likelihood under a model is averaged over the parameters of that model.

Figure

Baye's factor is part of the posterior odds

- Posterior odds
 - Relative probability for two models
 - Baye's factor
 - The part of posterior odds that depends on data
 - The ratio of the likelihoods
- Posterior odds: $\frac{p(m_1 | y)}{p(m_0 | y)} = \frac{p(y | m_1)}{p(y | m_0)} \frac{p(m_1)}{p(m_0)}$
- = Baye's Factor · Prior Odds

Likelihood under a model is averaged over the parameters

$$p(y | m) = \int d\theta p(y | \theta, m) p(\theta | m)$$

Figure 4: Slide 20 — Baye's factor is part of the posterior odds

6.4 — ROPE and HDI test

This section motivates interval null hypotheses by arguing that exact point-null models can be unrealistic. The ROPE defines a range of parameter values that count as practically equivalent to the null. The HDI/ROPE test then compares the posterior interval to this practical-null region.

New Definitions and Terms

- **ROPE** — region of practical equivalence; parameter values where the null model is practically true
- **Practical equivalence** — a situation where an effect may not be exactly zero but is too small to matter
- **HDI/ROPE test** — a hypothesis test based on the overlap between the posterior HDI and the ROPE
- **No decision** — the result when the posterior interval and ROPE overlap without one clearly containing the other

Formulas

$$p(\theta \in ROPE) < \alpha$$

If the posterior mass in the ROPE is small enough, the null model is rejected.

Figure

There are many possibilities for HDI / ROPE overlap

- No overlap
 - Reject null
- HDI in ROPE
 - Accept null
- All other possibilities
 - No decision

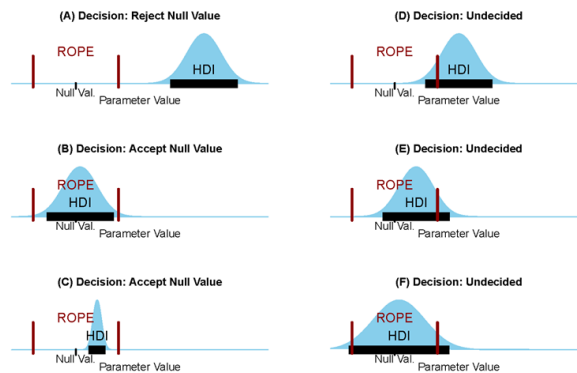


Figure 5: Slide 28 — There are many possibilities for HDI / ROPE overlap

6.5 — Significance

This section introduces significance from the perspective of type I error. A significant result is not just a posterior statement about one dataset; it is tied to the long-run probability of falsely rejecting a true null model. The lecture uses repeated simulated datasets under the null to ask whether the HDI/ROPE test is calibrated, conservative, or liberal.

New Definitions and Terms

- **Type I error** — falsely rejecting a true null model
- **error** — another name for type I error
- **Significance** — the probability of type I error for a hypothesis test
- **Significance level** — the chosen threshold α used to control type I error
- **Calibrated test** — a test whose type I error rate matches the target significance level
- **Conservative test** — a test with type I error probability smaller than α
- **Liberal test** — a test with type I error probability larger than α
- **Equivalence** — evidence that the parameter lies inside the ROPE

Formulas

$$p(\text{type I error}) < \alpha$$

A significance threshold controls the probability of falsely rejecting a true null model.

$$p(\theta \in \text{ROPE}) > \alpha$$

Equivalence means that enough posterior mass is inside the ROPE.

Figure

Our HDI / ROPE test is conservative

- Sampling multiple datasets
 - With true $\theta \in ROPE$
 - We find:
 - $p(HDI \cap ROPE = \emptyset) \approx 0.01$
 - Note: we used a strong prior!
- A hypothesis test is:
 - Conservative when $p(\text{type I error}) < \alpha$
 - Liberal when $p(\text{type I error}) > \alpha$

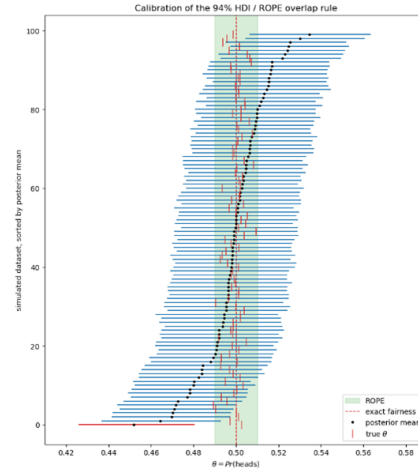


Figure 6: Slide 35 — Our HDI / ROPE test is conservative

6.6 — Basics of statistical reporting

This section shifts from analysis to reporting. Statistical reporting depends on the expectations of the context: papers, grants, pre-registrations, ethical approval, medical reporting, engineering applications, and patents may all have different norms or requirements. At the most basic level, the report should say what was measured, what was estimated, how uncertain the estimate is, what the effect size is, and whether the effect is significant.

New Definitions and Terms

- **Statistical reporting** — communicating the measurements, estimates, uncertainty, effect sizes, and hypothesis-test results expected in a scientific or applied context
- **Reporting standard** — a formal rule or informal norm for what statistical information must be reported
- **Parameter estimate** — the reported estimate of the model parameter or quantity of interest

Figure

The most basic level of reporting

- What did you measure?
- What is your parameter estimate?
- How is your interval?
- What is the effect size?
- Is the effect significant?

N tosses with probability of landing on heads α . Specifically, we obtained 175,421 heads out of 350,757 tosses, $\Pr(\text{heads}) = 0.5001$, 95% CI [0.4985, 0.5018]. Replacing the DHM same-side bias prior distribution with a highly informed Beta(5000, 5000) prior distribution (representing the background knowledge that in case a heads or tails bias exists, the effect is likely to be very small) yields moderate evidence against the presence of a heads-tails bias, $\text{BF}_{\text{heads-tails bias}} = 0.168$ (see Table 2 for by-coin summary).

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Figure 7: Slide 41 — The most basic level of reporting

6.7 — Power

This section introduces type II error and statistical power. Power is the probability that a test will identify a real effect when the null model is false. The lecture emphasizes three determinants of power: the true effect size, the amount of data, and the threshold used for significance.

New Definitions and Terms

- **Type II error** — failing to reject a false null model
- **error** — another name for type II error
- **Power** — the probability of successfully identifying a true effect
- **More powerful test** — a test with a smaller probability of type II error

Formulas

$$\text{power} = 1 - \beta$$

Power increases when the probability of type II error decreases.

Figure

Power depends on effect size, sample size, and significance threshold

Power is the probability of successfully identifying a true effect

Three things affect power

1. The true effect size
 - Larger effects $\rightarrow$ HDI farther from ROPE $\rightarrow$ easier to reach significance
2. The amount of data
 - More data $\rightarrow$ smaller HDI $\rightarrow$ easier to reach significance
3. The threshold for significance
 - ~~Smaller~~ $\alpha \rightarrow$ smaller HDI $\rightarrow$ easier to reach significance

larger α

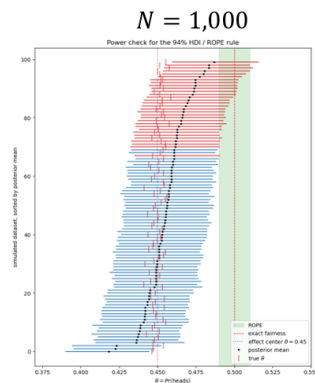
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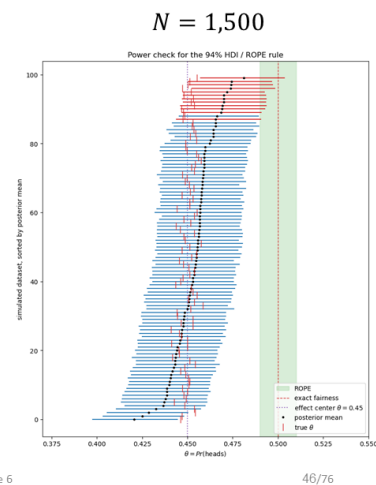
Figure 8: Slide 44 — Power depends on effect size, sample size, and significance threshold

Power under increased N

- Increasing N to 1,500
 - Power increases to 0.9



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Figure 9: Slide 46 — Power under increased N

6.8 — Pre-hoc and post-hoc power

This section distinguishes two uses of power analysis. Pre-hoc power analysis asks how much data are needed to reach a target power for an expected or minimally meaningful effect. Post-hoc power analysis asks, after the experiment has already been run, how sensitive the experiment was and how large an effect would have needed to be to reach the target power.

New Definitions and Terms

- **Pre-hoc power analysis** — power analysis used before collecting data to choose an appropriate sample size
- **Post-hoc power analysis** — power analysis used after collecting data to evaluate the sensitivity of the completed experiment
- **Minimal effect size** — the smallest effect that would be meaningful enough to detect
- **Sensitivity** — the size of effect an experiment could reliably identify

Formulas

$$\text{minimal effect size} = \frac{1}{2} \times \text{small effect size}$$

The lecture gives this as a rule of thumb for choosing a minimal effect size.

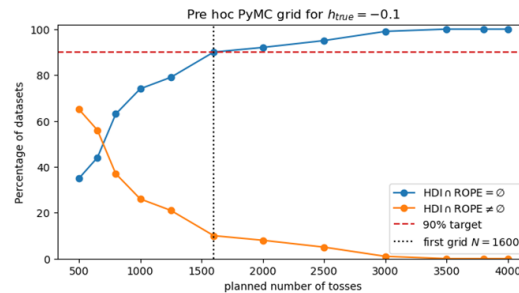
$$ROPE = \frac{1}{2} \times \text{minimal effect size}$$

The lecture gives this as a rule of thumb for choosing the ROPE.

Figure

Pre-hoc power: choosing N to get enough power

- How big a sample do we need?
- We want a power of 0.9
 - For very small effects
 - Cohen's $h = 0.1$
- How big a dataset do we need?
- Simulate!
 - Measure power
 - As a function of N
 - Find N
 - Where power > 0.9
 - $N \approx 1,600$



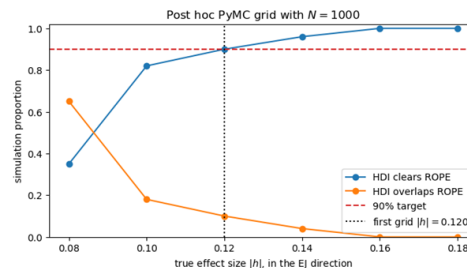
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Figure 10: Slide 49 — Pre-hoc power: choosing N to get enough power

Post-hoc power: how big an effect do we need?

- We've done the experiment
 - N is fixed
 - $\hat{\theta}$ is known
- We want to know how sensitive our experiment is
 - If the effect had been small
 - Would it be significant?
- Simulate!
 - Measure power
 - As a function of effect size
 - Find smallest effect size
 - Where power > 0.9



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Figure 11: Slide 52 — Post-hoc power: how big an effect do we need?

6.9 — Frequentist methods

This section introduces the frequentist hypothesis-test structure using a statistic whose sampling distribution is known under the null hypothesis. The test asks whether the statistic calculated from the observed data falls in the tail of that null sampling distribution. The lecture then compares frequentist tests with Bayesian HDI/ROPE tests and emphasizes that different tests can disagree.

New Definitions and Terms

- **Statistic** — a quantity calculated from data and used for inference
- **Sampling distribution** — the distribution of a statistic across repeated datasets generated under a model
- **z statistic** — a standardized statistic that is approximately normally distributed under the null when the sample size is large
- **p value** — the probability, under the null model, of getting data as extreme as the observed data or more extreme
- **Null sampling distribution** — the distribution of the statistic when the null model is true

Formulas

$$\hat{\theta} = \frac{y}{N}$$

The observed proportion is the number of successes divided by the total number of trials.

$$z = \frac{\hat{\theta} - \theta_0}{\sqrt{\frac{1}{N}\theta_0(1 - \theta_0)}}$$

For large N under the null hypothesis, the z statistic is approximately normally distributed.

$$z \sim N(0, 1)$$

The normal approximation allows the observed statistic to be compared with the null sampling distribution.

Figure

Frequentist hypothesis tests use a statistic

- Find a statistic
 - Whose sampling distribution is known
 - Under the null hypothesis
 - Test whether calculating the statistic on our data is consistent
 - With the null sampling distribution
 - That means:
 - Is measured statistic
 - Is in tail
 - Of null sampling distribution

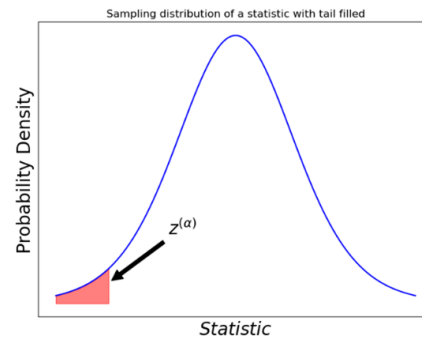


Figure 12: Slide 55 — Frequentist hypothesis tests use a statistic

Different hypothesis tests don't always agree

- TianqiPeng was significant for z test and HDI / ROPE
- EJ was not significant for either z test or HDI / ROPE
- DavidV was significant for z test but not for HDI / ROPE

	person	coin	N	heads proportion	difference from 0.5	z statistic	normal approx p-value
0	EJ	50CZK	200	0.440	-0.060	-1.70	0.08969
1	TianqiPeng	0.20EUR	1000	0.445	-0.055	-3.48	0.0005042
2	davidV	0.10EUR	3660	0.480	-0.020	-2.38	0.0173

Figure 13: Slide 58 — Different hypothesis tests don't always agree

6.10 — Comparing two groups

This section applies the lecture's ideas to comparing outcomes in hospitals with and without PCI capabilities. The hierarchical model from the previous lecture is extended by adding deterministic quantities for the PCI and non-PCI means and their difference. The effect sizes are very small, the HDI overlaps the ROPE, and the lecture uses both Bayesian and frequentist approaches to think about effect size, significance, and sensitivity.

New Definitions and Terms

- **PCI** — percutaneous coronary intervention, used here to define one hospital group
- **Deterministic parameter** — a model quantity calculated from other parameters and stored in the `idata` object
- **Group comparison** — estimating and testing a difference between two groups
- **Difference parameter** — the parameter representing the contrast between PCI and non-PCI groups

Formulas

```
mu_nonpci = pm.Deterministic("mu_nonpci", mu_type[0])
mu_pci = pm.Deterministic("mu_pci", mu_type[1])
mu_difference = pm.Deterministic("mu_difference", mu_pci - mu_nonpci)
```

Deterministic parameters store calculated quantities of interest in the inference data object.

$$\hat{\theta}_i = \frac{y_i}{N_i}, \quad i = 1, 2$$

Each group has its own observed proportion.

$$\hat{\theta}_{12} = \frac{y_1 + y_2}{N_1 + N_2}$$

The pooled estimate is used in the frequentist normal approximation for comparing two groups.

$$\frac{1}{N_{12}} = \frac{1}{N_1} + \frac{1}{N_2}$$

N_{12} combines the uncertainty from the two group estimates.

$$z = \frac{\hat{\theta}_1 - \hat{\theta}_2}{\sqrt{\frac{\hat{\theta}_{12}(1-\hat{\theta}_{12})}{N_{12}}}}$$

The two-group frequentist test is based on the same normal-approximation logic introduced for one proportion.

Figure

Extend the model to calculate the difference

- We can use calculations inside the model to get quantities of interest

```
mu_nonpci = pm.Deterministic("mu_nonpci", mu_type[0])
mu_pci = pm.Deterministic("mu_pci", mu_type[1])
mu_difference = pm.Deterministic("mu_difference", mu_pci - mu_nonpci)
```

Deterministic parameters go into the idata object

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_nonpci	0.084	0.009	0.067	0.102	0.000	0.000	4012.0	3026.0	1.0
mu_pci	0.076	0.006	0.065	0.088	0.000	0.000	5166.0	3238.0	1.0
mu_difference	-0.008	0.011	-0.028	0.013	0.000	0.000	4198.0	2989.0	1.0

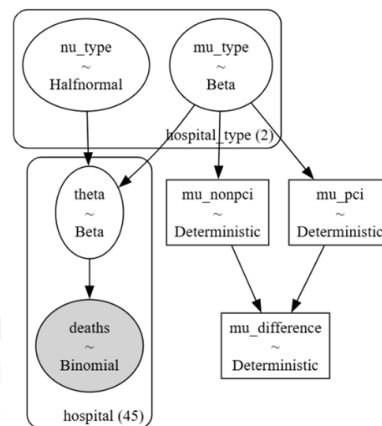


Figure 14: Slide 65 — Extend the model to calculate the difference

Comparing HDI to ROPE does not allow conclusions

- HDI intersects with ROPE
- ROPE does not contain HDI

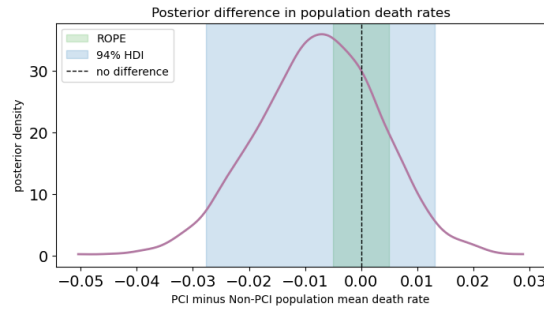


Figure 15: Slide 67 — Comparing HDI to ROPE does not allow conclusions

Frequentist hypothesis test for comparison of two groups

- Based on same normal approximation

$$\hat{\theta}_i = \frac{y_i}{N_i} \quad i=1,2$$

$$\hat{\theta}_{12} = \frac{y_1 + y_2}{N_1 + N_2}$$

$$\frac{1}{N_{12}} = \frac{1}{N_1} + \frac{1}{N_2}$$

$$z = \frac{\hat{\theta}_1 - \hat{\theta}_2}{\sqrt{\hat{\theta}_{12}(1 - \hat{\theta}_{12}) \frac{1}{N_{12}}}} = -1.32$$

Figure 16: Slide 70 — Frequentist hypothesis test for comparison of two groups